

PROJECT PORTFOLIO

Smart RFID Automated Gate System

Prepared by

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Diploma in Computer Science · Class: M3CS1104C

This project was completed as a group effort.

Prepared for:

Sir Alias bin Deraman

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1. PROJECT OVERVIEW

This portfolio documents the Smart RFID Automated Gate System, developed as the final group project for ITT270 (Digital Electronics) during Semester 4 at UiTM Jasin. While produced as a collaborative group effort, this portfolio is submitted by Adam Iskandar bin Nazri (2024277906) as an individual academic record of the project work.

The project integrates hardware automation with multi-factor authentication to produce a secure, energy-efficient automated gate system suitable for modern residential IoT deployment.

Course	ITT270 — Digital Electronics
Class	M3CS1104C
Project Title	Smart RFID Automated Gate System
Platform	Arduino Uno (C++ / IoT)
Key Technologies	RFID, Ultrasonic Sensor, LDR, Keypad, Servo Motor, LCD
Submitted by	Adam Iskandar bin Nazri (2024277906)
Project Type	Group Project
Lecturer	Sir Alias bin Deraman

2. INTRODUCTION

The rapid evolution of smart home technology has transformed residential security, moving properties away from the vulnerabilities of physical keys and standard remote controls. Legacy entry systems create single points of failure — a lost key or remote fob locks out residents and exposes properties. They also provide no independent access method for expected visitors or emergency services, causing unnecessary delays.

Additionally, conventional outdoor security lighting operates on inefficient timers or requires manual operation, resulting in dark, hazardous driveways at night or wasted energy during daylight hours.

To address these issues, this project presents the Smart RFID & Keypad Automated Gate System with Night-Light Integration — a dual-authentication framework combining RFID contactless access for residents with a 4x4 matrix keypad PIN override for guests. An ultrasonic sensor keeps the system in low-power standby until a vehicle is detected, while a Photoresistor (LDR) automatically triggers driveway lighting in dark conditions.

3. PROBLEM STATEMENT

Modern residential perimeters frequently suffer from security gaps and energy inefficiencies due to rigid, non-automated entry systems. Three core problems were identified:

- Single point of failure: Physical keys or remote fobs can be lost or forgotten, locking out residents and leaving properties exposed.
- No guest access: Legacy systems offer no secure, independent access method for visitors, couriers, or emergency services, causing unnecessary delays.
- Inefficient lighting: Conventional outdoor security lighting uses simple timers or stays manual, causing dark hazardous driveways at night or unnecessary energy waste during the day.

A smart, automated gate entry system featuring dual-factor authentication and responsive, energy-saving environmental lighting addresses all three issues.

System Behaviour Summary

Car At Gate	Authorized RFID / PIN	Ambient Light (LDR)	Gate Servo	Light
Yes	Yes	Dark (Night)	OPEN	ON
Yes	Yes	Bright (Day)	OPEN	OFF
Yes	No	—	LOCKED	OFF
No	—	—	LOCKED	OFF

4. LOGIC DESIGN

4.1 Truth Table

Three inputs (CAR, AUTH, LDR) determine two outputs (GATE and LIGHT). The truth table below enumerates all eight input combinations:

Inputs:

Variable	1 = True Condition	0 = False Condition
CAR (Ultrasonic)	1 = Car Present	0 = No Car
AUTH (RFID / PIN)	1 = Authorized	0 = Unauthorized / None
LDR (Light Sensor)	1 = Dark (Night)	0 = Bright (Day)

Outputs:

Variable	1 = Active State	0 = Inactive State
GATE (Servo)	1 = Open	0 = Locked
LIGHT (Driveway LED)	1 = On	0 = Off

INPUT			OUTPUT	
CAR (A)	AUTH (B)	LDR (C)	GATE	LIGHT
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1

Figure 1 — Full truth table for the Smart Gate system

4.2 Karnaugh Maps

Karnaugh maps were used to derive the minimal Boolean expressions for each output:

- $GATE = A \cdot B$ (gate opens only when a car is present AND authentication is successful; light conditions are irrelevant)
- $LIGHT = A \cdot C$ (driveway light activates only when a car is present AND ambient light is dark; authentication status is irrelevant)

1. K-Map for GATE (Servo)

The gate only opens if a car is present (A) AND the user is authorized (B), regardless of whether it is day or night (C).

	A'B'	A'B	AB	AB'
C'	0	0	1	0
C	0	0	1	0

Grouping: A vertical pair in the **AB** column. (The C variable drops out).

Equation: $GATE = AB$

2. K-Map for LIGHT (Driveway LED)

The driveway light only turns on if a car is present (A) AND it is dark out (C), regardless of whether they have scanned a valid tag yet (B).

	A'B'	A'B	AB	AB'
C'	0	0	0	0
C	0	0	1	1

Grouping: A horizontal pair in the bottom row (C) spanning across columns **AB** and **AB'**. (The B variable drops out).

Equation: $LIGHT = AC$

Figure 2 — Karnaugh maps for GATE and LIGHT outputs

4.3 Logic Circuit

The simplified Boolean expressions translate directly into two AND gates — one for the gate servo and one for the driveway LED — sharing the Car Present (A) input line.

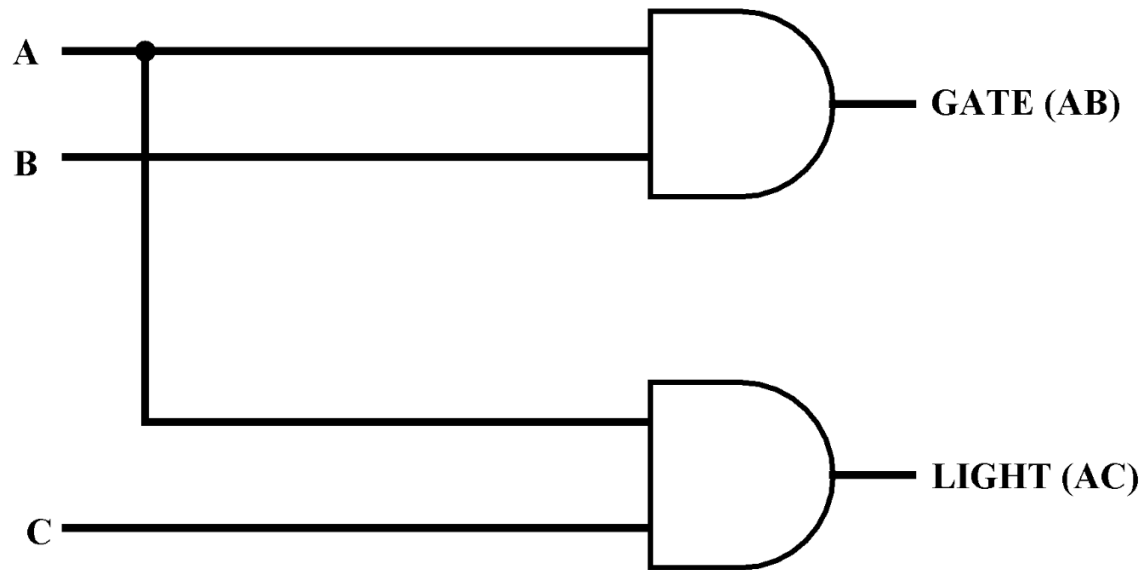


Figure 3 — Logic circuit diagram ($GATE = AB$, $LIGHT = AC$)

5. IoT SYSTEM DESIGN

5.1 Components

The system is governed by an Arduino Uno microcontroller and integrates the following hardware:

- Arduino Uno — Central processing unit executing control logic for gate and lighting subsystems
- MFRC522 RFID Reader — Primary resident authentication via contactless RFID tags
- HC-SR04 Ultrasonic Sensor — Measures vehicle proximity to trigger "Car Present" logic
- 4x4 Membrane Keypad — Secondary authentication via secure 4-digit PIN for guests
- Photoresistor (LDR) — Monitors ambient light to enable automated energy-saving lighting
- Servo Motor — Gate actuator; opens on valid authentication
- LCD 1602 (I2C) — Displays real-time status feedback
- LED Indicators — Red (denied), Green (granted), White (driveway light)

5.2 IoT Diagram

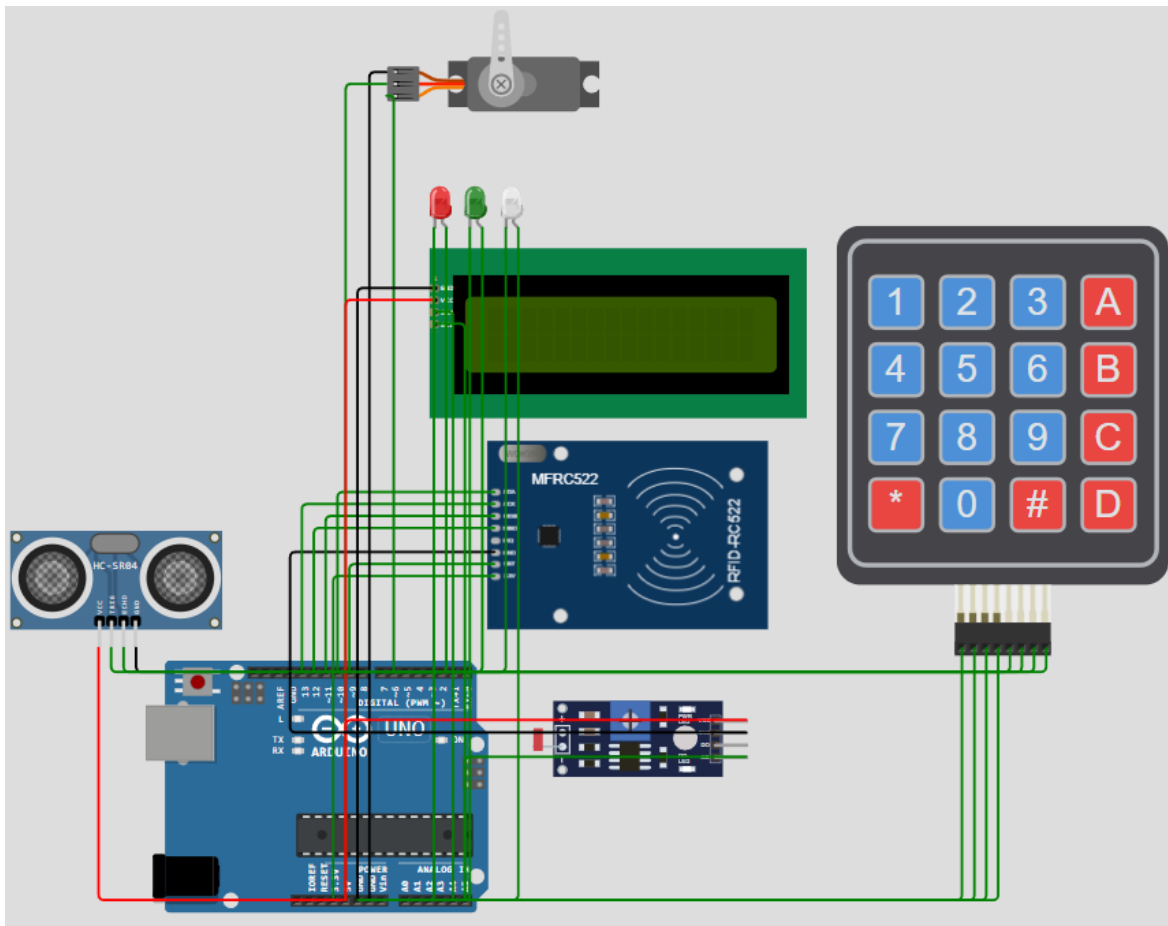


Figure 4 — Wiring diagram of the full IoT system

6. SYSTEM OPERATION (SCREENSHOTS)

The following screenshots from the Wokwi simulation demonstrate each operating state of the Smart Gate System:

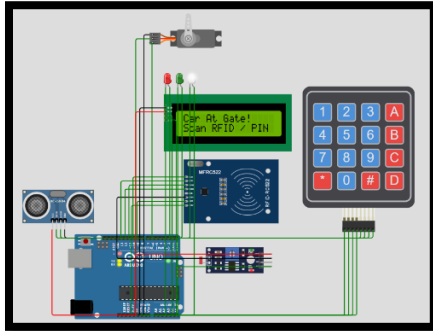


Fig. 5 — Car detected at gate; LCD prompts for RFID or PIN

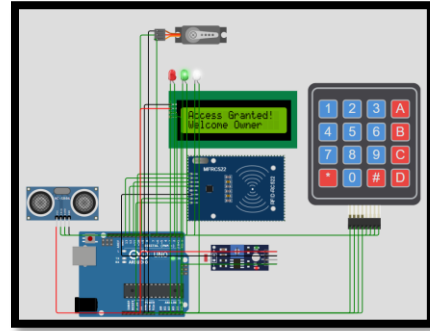


Fig. 6 — Owner RFID card scanned; access granted

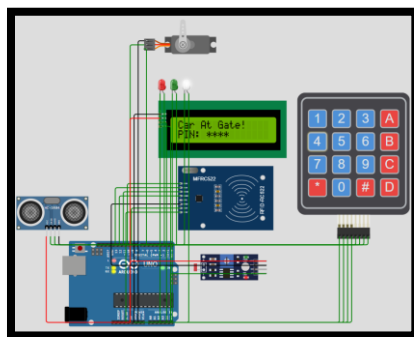


Fig. 7 — VIP guest entering 4-digit PIN on keypad

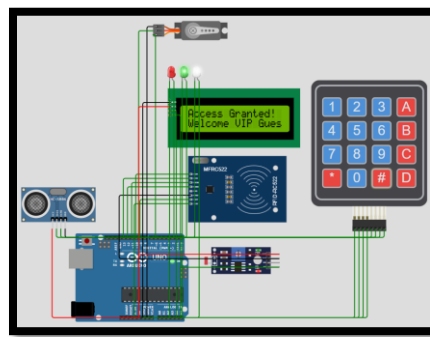


Fig. 8 — Correct PIN entered; VIP access granted

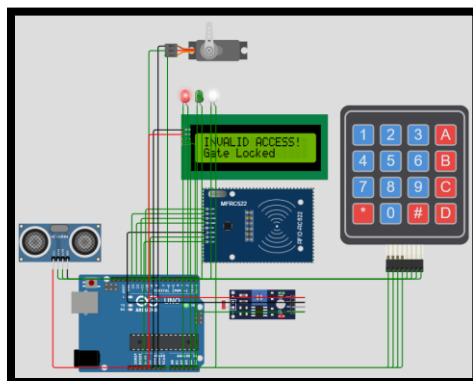


Fig. 9 — Invalid card or wrong PIN; gate remains locked and red LED activates

7. SKILLS DEMONSTRATED

- Digital logic design — Boolean algebra, truth tables, Karnaugh map minimisation
- Circuit design — AND gate implementation, multi-output combinational circuits
- Arduino programming (C++) — sensor polling, conditional logic, actuator control
- IoT integration — RFID (MFRC522), ultrasonic (HC-SR04), LDR, keypad, servo, LCD
- Dual-factor authentication — RFID UID matching + PIN validation logic
- Energy-efficient design — LDR-driven conditional lighting (LIGHT = AC)
- System simulation — Wokwi virtual prototyping and circuit validation
- Technical documentation — Truth tables, K-maps, circuit diagrams, IoT wiring

8. ABOUT THE STUDENT

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Group Members (Collaborative Project)

This project was completed as a group effort. The other members of the team were:

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9. REFERENCES

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